ISMRM & ISMRT
ANNUAL MEETING & EXHIBITION

Precision Functional Gradients via Optimal Shrinkage Denoising

Individual gradients with fewer voxel needed

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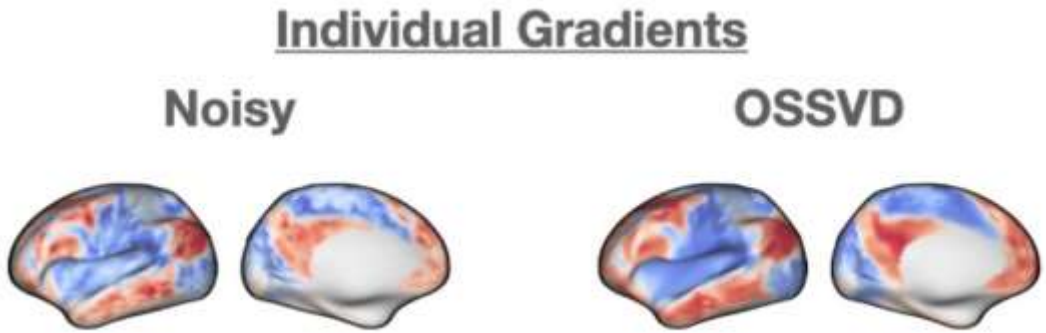

Declaration of Financial Interests or Relationships

Speaker Name: Jiarui Mao

I have no financial interests or relationships to disclose with regard to the subject matter of this presentation.

Motivation

- Resting-state fMRI gradients map functional hierarchies in the neocortex
- High noise limits analysis to group-level; individual gradients remain underexplored



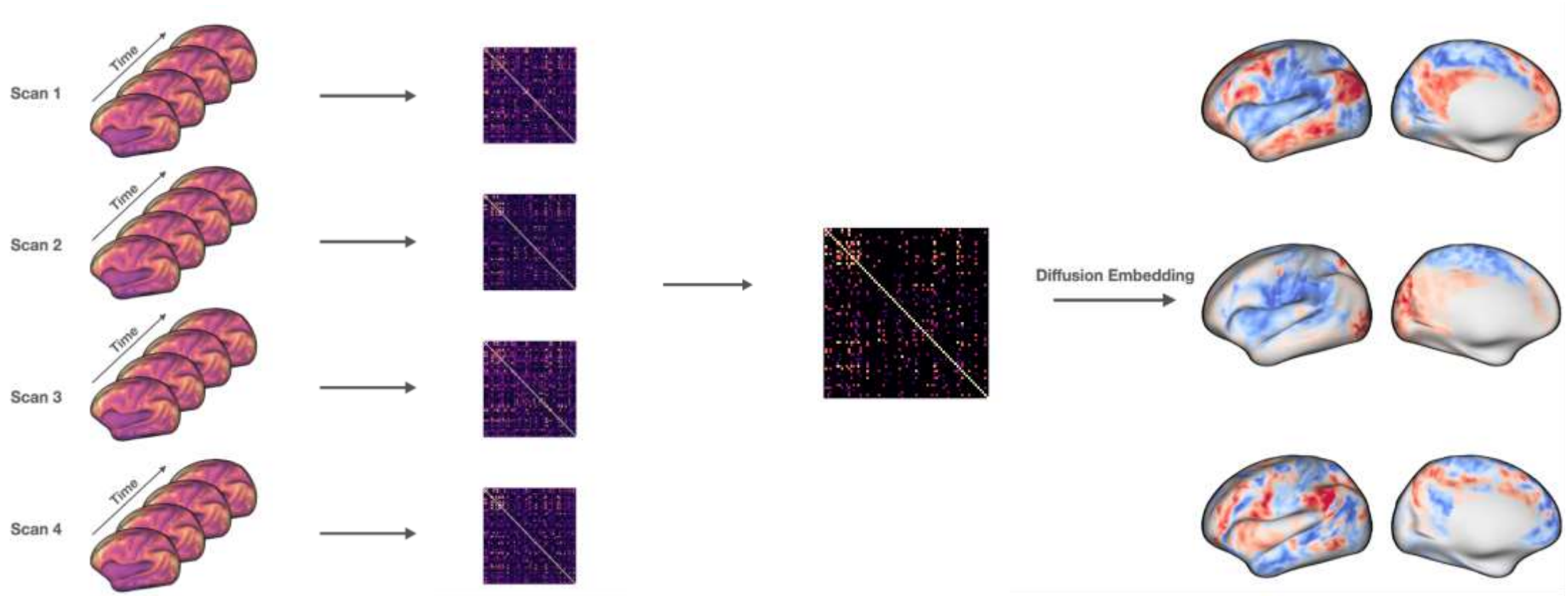
Methods

Data and denoising

- BCP and HCPYA datasets
- compared with other denoising methods: MP-PCA, NORDIC

Methods

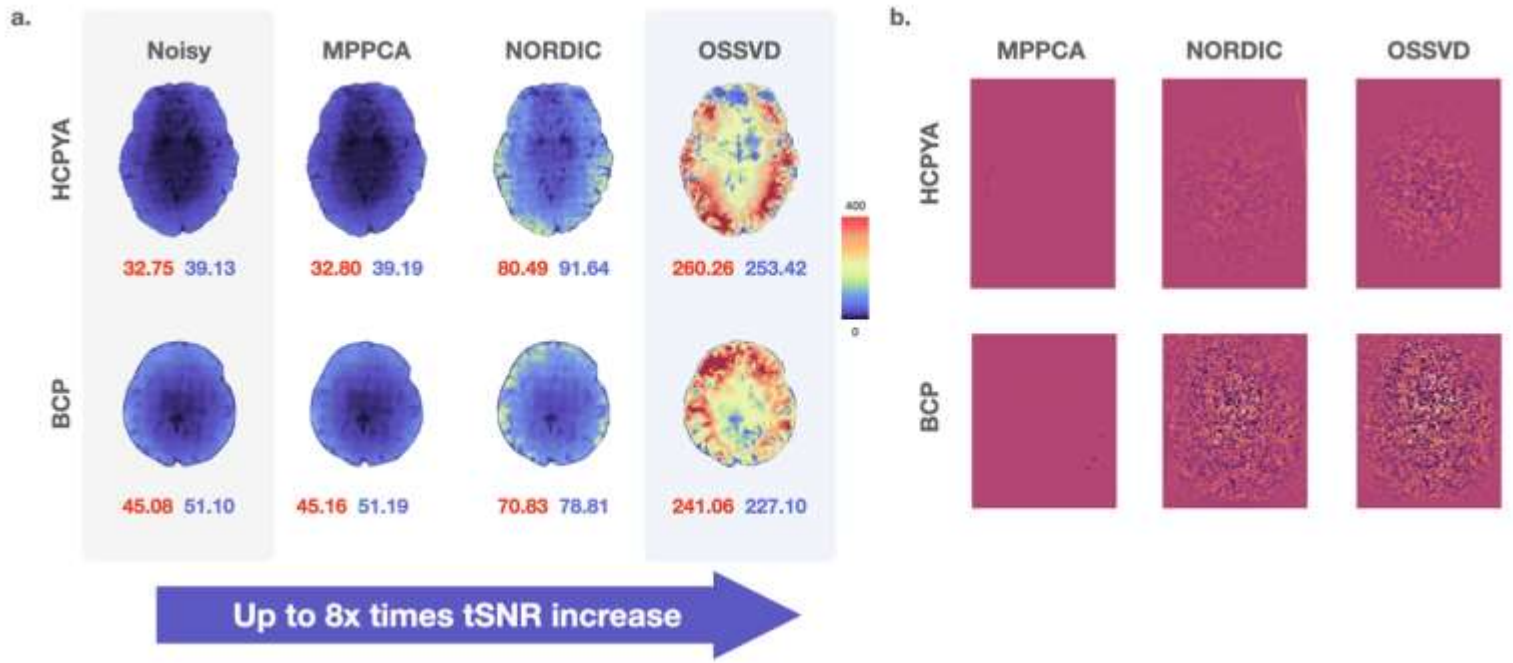
Individual gradient computation



Results

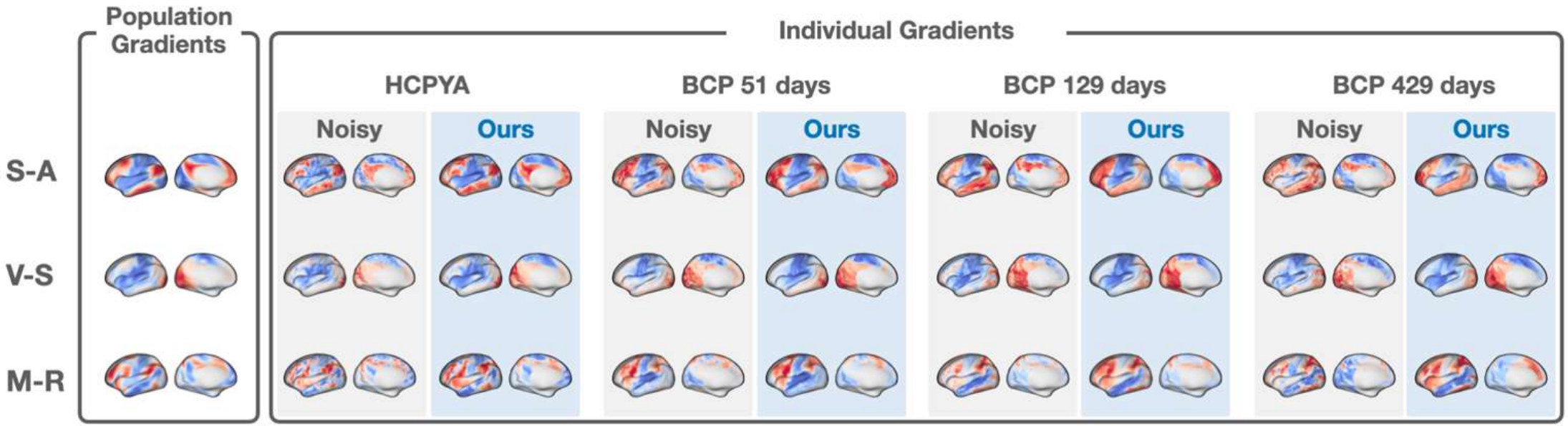
tSNR increases

- outperform MPPCA by 800%
- outperform NORDIC by 300%
- residual maps: no discernible structural features



Results

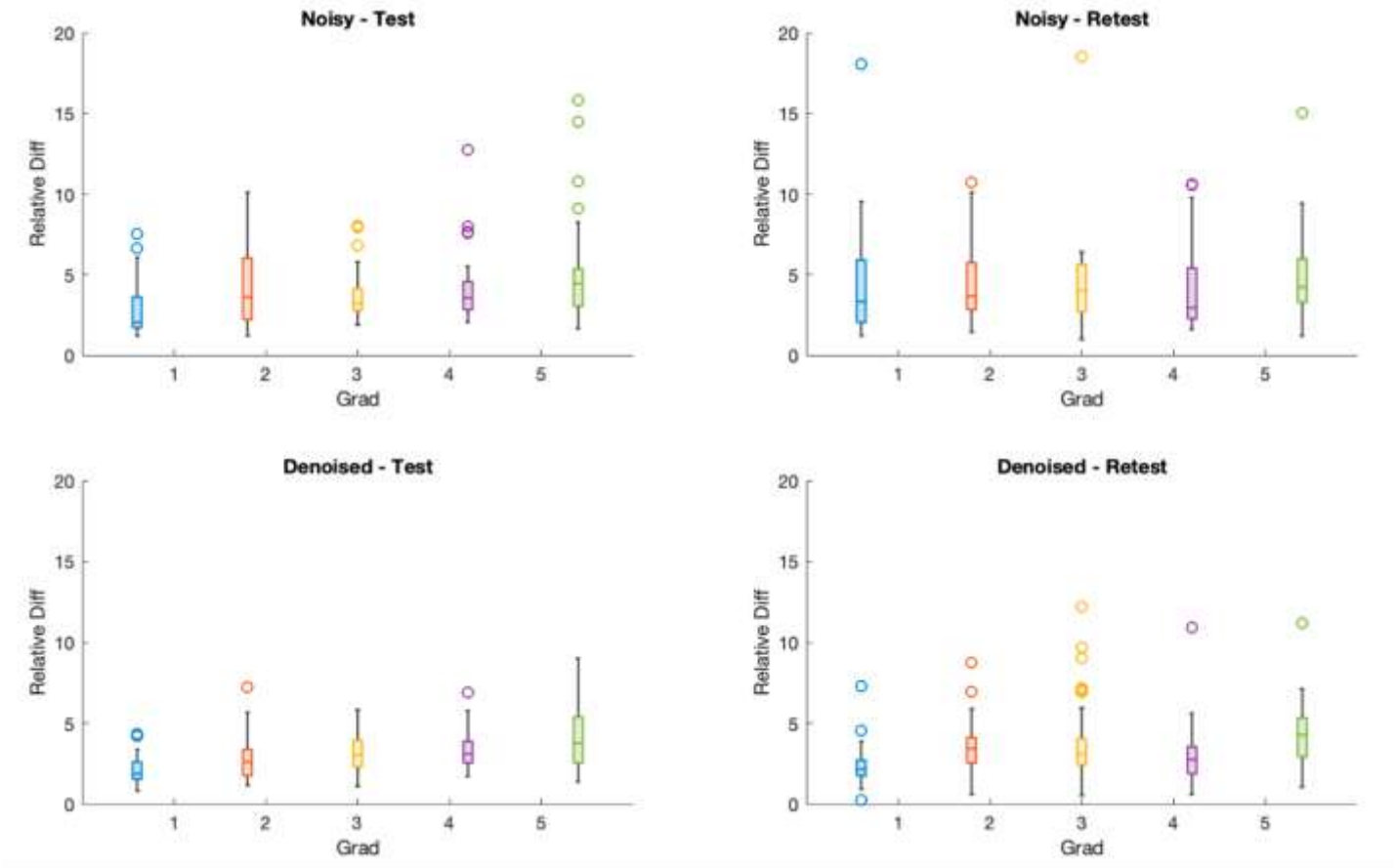
Improvement in Quality Individual Gradients



Results

Decrease in relative difference compared with reference

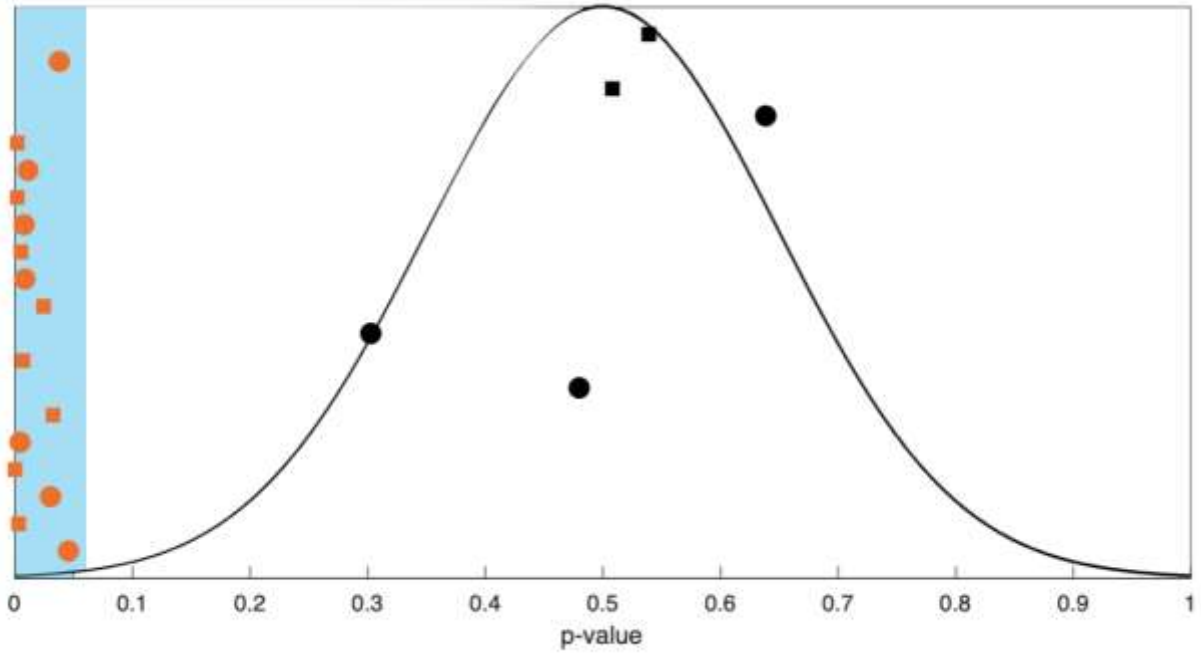
- reference = 1200 noisy test(more)
- 300
- 10 subjects



Results

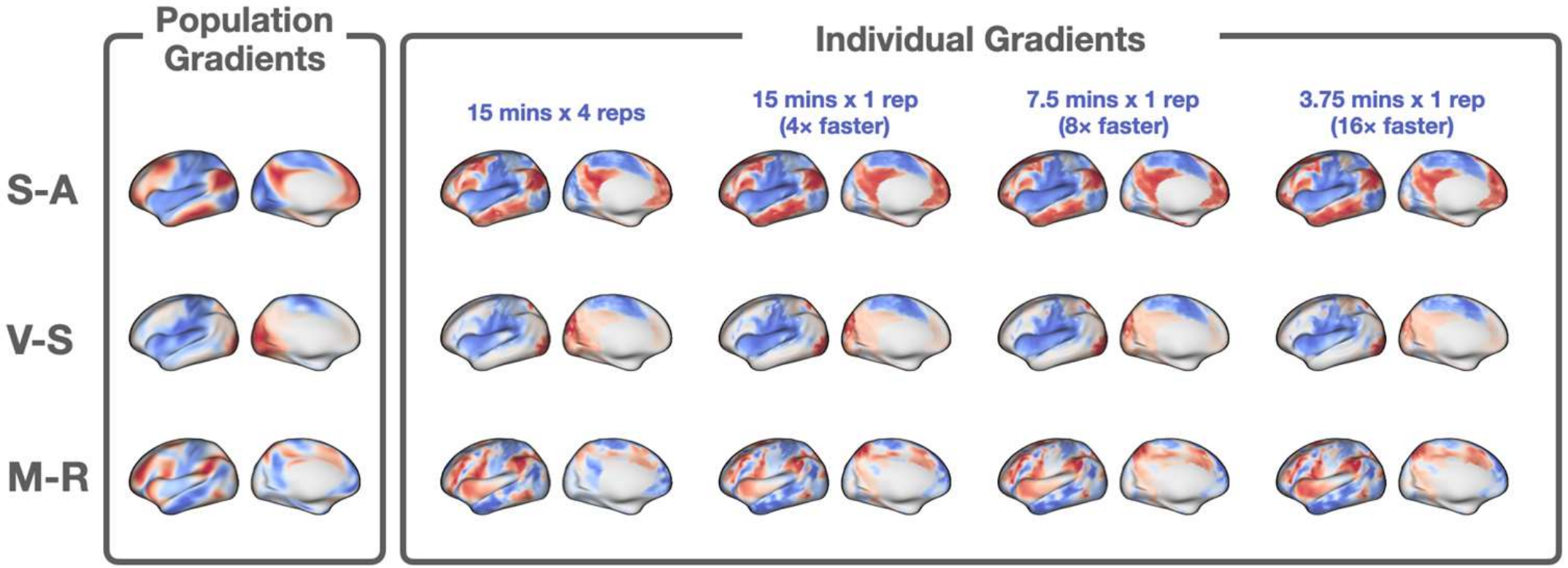
Significant decrease

- t-test shown 75% positive rate
- possible outliers



Conclusions

individual gradients



- able to compute individual gradients with